CSE 4511 : Computer Networks

Course Outline

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Office: Room 503, AB2 Building
Semester: Winter 2025
Lectures: Mon/Wed: Room number - 301, 302 & 105, 106 (AB3)
Pre-requisite: CSE 4405 (Data & Telecommunication)

Contents

1. Syllabus

Course Format

The course will meet twice a week for 75-minute class meetings.

Each class meeting will contain discussion, activities, and material that assumes that you have watched the preparation videos/book contents in advance.

There is also a lab associated with this course. The lab content will be given separately.

Course Contents

Introduction: Internet, Network edge, Network Core, Access Networks, Delay, Loss, and Throughput in Packet-Switched Networks, Protocol Layers, and Service models, Security

Underlying technologies: Wired LANs, Wireless LANs

Addressing, ARP, Switches, VLANs, VPNs & Link Virtualization

Network layer: IP addressing, IP packet forwarding, Subnetting, CIDR, Internet protocol, ICMP, ARP, DHCP, and IPv6 overview; Routing protocols

Transport layer: Introduction to transport layer, UDP, TCP, SCTP

Application layer: Principles of Network applications, Web & HTTP, Electronic mail, DNS

Network Security

Reasoning Behind the Course Contents

This course is a continuation of previous semester course - Data and Telecommunication, where the students have completed the fundamentals about how the communications take place in a network, between the networking devices. In this course, we are going to look into what is the content of those communications. At first, we are going to talk a high level overview of how the network communication we know today works. Then we are going to examine the different components of a network communication from a engineering perspective. We are going to see how the raw bits that we send from a networking device end up becoming meaningful layer by layer. The main objective of this course relies on, making the students critically sound to design/re-design any component of computer networking, when needed in real life.

Assessment Breakdown

Midterm exam	25%
Final exam	50%
Quizzes / Assignments	15%
Attendance	10%

Grading Scale

Letter Grade	Percentage Range (%)
A+	80–100
A	75–79
A-	70–74
B+	65–69
B	60–64
B-	55–59
C+	50–54
C	45–49
D	40–44
F	<40

Table 1: Grading Scale

Late Submission Policy

- Each student begins the term with a **96-hour grace period**.
- **Assignments are due at 11:59 p.m.**
- Every hour (or part of an hour) late deducts one hour from your grace balance.
- While grace hours remain, late submissions incur **no penalty**.
- Once the grace period reaches **zero**:
  - Work submitted within one week receives **half credit**.
  - Work submitted more than one week late receives **no credit**.
- **Medical emergencies** with valid documentation are exempt. Other reasons (e.g., interviews, conferences, travel) are not.

Attendance Policy

While regular attendance is strongly encouraged, the primary goal of this course is effective learning and performance. Students who demonstrate strong understanding through assessments will not be penalized for occasional absences. Specifically, **if a student achieves 80% or higher in a quiz (subject to change), they will receive full attendance credit for the class sessions covering the topics assessed in that exam, regardless of actual attendance on those days**.

Reading and Videos

Supplementary Videos:

Prof. Feamster's Networking Videos

Supplementary Textbooks:

1. Computer Networking: A Top-Down Approach (8th edition), by Jim Kurose and Keith Ross. PDF Link
2. Data Communication and Networking by Behrouz A Forouzan. PDF Link
3. TCP/IP Protocol Suite (4th Edition), by Behrouz A. Forouzan. PDF Link

2. Schedule

Overview

This schedule and syllabus is preliminary and subject to change. Edits will be ongoing, but you can assume that all updates for the subsequent week will be completed by 5 PM of the Sunday prior to the week of instruction.

Week	Topic	Reading	Exercise
1.1	**Introduction & Revision**		
1.2, 2.1	**Introduction:** Internet, Network edge, Network Core, Access Networks, Delay, Loss, and Throughput in Packet-Switched Networks, Protocol Layers, and Service models	KR: 1.1-1.4, Slide Link	KR (Chapter 1): P1-17, P25-30
2.2	**Data link layer:** Underlying technologies: Wired LANs, Wireless LANs	TCP/IP: 3, Slide Link	TCP/IP (Chapter 3): E1-7
3, 4.1	**Network layer:** Logical Addressing (with IPv6), NAT, DHCP	DCBA: 19, Slide Link 1, DHCP, Ip Address, Subnetting	KR (Chapter 4): P11,14,15, DCBA (Chapter 19): Full
4.2, 5.1	**Network layer:** Routing Algorithms & Protocols	KR: 5.2-5.4 , Slide Link	KR (Chapter 5): P3-P15
5.2, 6.1	**Data link layer:** Addressing, ARP, Switches, VLANs, VPNs	KR: 6.4.1, 6.4.3, 6.4.4, 6.5, Slide Link	KR (Chapter 6): P14,15,16
6.2, 7	**Network layer:** Internet protocol v4, ICMPv4	TCP/IP: 7,9 (Without Package), Slide Link: 7, 9	TCP/IP: 7,9
8	**Network layer:** Internet protocol v6	TCP/IP: 27 (Without Package), Slide Link	TCP/IP: 27
11	**Mid Solution Discussion & Showing Script**	–	
12	**Transport layer:** Introduction to transport layer, UDP	TCP/IP: 13, 14.1-14.4, Slide Link: 13, 14	To be announced
13, 14.1	**Transport layer:** TCP	TCP/IP: 15.1-15.9, Slide Link	To be announced
14.2, 15	**Transport layer:** SCTP	TCP/IP: 16.1-16.9, Slide Link	To be announced
16	**Application layer:** Principles of Network applications, Web & HTTP, Electronic mail, DNS	KR: 2.1-2.4, Slide Link	KR(Chapter 2): P1-P21
16.1	**Overview & Suggestions**	–	

DCBA: Data Communication Behrouz A. (4th edition), KR: Kurose/Ross (8th edition), TCP/IP: TCP/IP (4th Edition) by Behrouz A.

3. Resources

Primary textbook

- **Computer Networking: A Top-Down Approach** — Kurose & Ross (8th edition).
- **Data Communication and Networking** - Behrouz A Forouzan (4th edition)

Online tools and tutorials

- Wireshark for packet capture and Analysis - Resource Link
- Introduction to Python Programming - Youtube Video Link
- Packet Tracer Tutorial Link
- Python Socket Programming Tutorial Link

Google Classroom

All lecture slides, assignment statements and grades will be posted on the course google classroom. The google classroom code is - **5v5lpwgd**

Course Feedback Form

Anonymous Course Feedback Form Link

Individual Result Link

Given in the classroom